

Project 2, Weather app

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This is the documentation of web development project, where the goal was to build a weather web application. The project uses few different api's to gain weather information, location information and icons to visualize the weather. Font Awesome icons were used in this project, as I have come familiar with them on previous course "Software development skills – Full stack". The code is in english, but the comments on the functions and the user interface is in finnish.

Declaration of AI usage

1. The following AI systems were used on this project: Claude AI and Google Gemini.
2. The mentioned AI systems were used as followed:
 - a) Asked from Claude AI, "what different ways are there to save users, forexample favorite weather location on weather web app". Then localStorage was the one I chose. Here is the function (also mentioned on the code)

```
/*Hae taan suosikit local storagesta (FUNKTION ON TEHNYT CLAUDE AI)*/  
function getFavorites() {  
  return JSON.parse(localStorage.getItem("favorites")) || [];  
}
```

- b) I ran into a problem with the formatting of openWeatherData, so this part is written by Claude AI:

```
/*Open weather */  
let openWeatherRows = "";  
const offset = openWeatherData.city.timezone;  
for (let i = 0; i<8; i++) {  
  const entry = openWeatherData.list[i];  
  const hour = new Date((entry.dt+offset)*1000).toISOString().slice(11, 16);  
  const isDay = entry.weather[0].icon.endsWith("d");  
  const icon = openIcon(entry.weather[0].id, isDay);  
}
```

c) Google gemini was used for the font awesome icons css:

```
/* Ikonit (GOOLE GEMINI)*/
td i.fa-solid, td svg.svg-inline--fa {
  font-size: 18px;
  width: 24px;
  text-align: center;
  color: #3498db;
}
```

The finished project works on both desktop and mobile devices. The site shows weather data by two different providers for 24 hours. Other shows with 1 hour steps and the other one on 3 hours steps. They both show time, temperature and what the weather is like with an icon.

It is possible for the user to search weather data on a specific location by using the search bar and typing in the name of the location or by pressing "paikanna" to show the weather of users current location. After some location has been searched, the site shows the current weather by one provider and 24 hour weather by two different. Then its also possible for the user to save the location as a favorite by pressing "Tallenna Suosikkeihin", to access the weather data on that location quicker next time.

The table below shows the implemented features and points proposal

Feature	Description	Points
Well written pdf report	This documentation	3
Responsivness	It works on both mobile and desktop	4
Application works on multiple browsers	-	3
The application has clear directory and structure.	It has clear directory and structure, and comment on code.	2
User can search for location	There is a search bar that takes text as an input	1
Geolocation API	User can use gps location to find local weather data	2
Two data providers	Project uses two different weather data providers	2
User sees current weather at specific location	The site shows a card of current weather on location user has searched	1

Users sees forecast for 24 hours / hourly based	One of the used data providers shows data 24 hours / hourly based	3
All the weather forecast elements uses icons (and numbers) for e.g. sunny and cloudy weathers	Temperature is shown by number on celsius and weather with font awesome icons	3
User has the option to tag some locations as her favorites and thus access them from the favorites menu	It is possible. Data is saved on localstorage	2
Use of reverse geoapi	Reverse geocoding made it possible to show the name of location, when user uses gps.	1
Total		28p